GROUP 13: RC CAR PROJECT MID-TERM REPORT

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ABSTRACT

The purpose of this project was to design, build, and race an RC car around a predetermined track. The race will not be a time trial, and we will be competing against other student teams. The race allows mechanical engineering students to gain experience and supplement their understanding of ME 338. Students own their projects, from design to the race, strengthening understanding of engineering principles. Currently, our RC car chassis is a CAD model with 3D-printed parts. In pre-testing, we calculated vehicle performance data, which will be adjusted per test results. We designed the entire chassis layout sans the steering system. Completion of the car's layout lets us make estimations of stress and deflection in our vehicle. We found that the chassis of our vehicle will yield in a head-on collision with a solid object at top speed. These findings will guide design improvements and testing to optimize durability and performance.

Keywords: 3D print, CAD, FEA, FOS, RC car, yield strength.

NOMENCLATURE

KE	Kinetic Energy
U	Elastic strain energy
E	Young's Modulus
G	Shear Modulus
A	Cross-sectional Area
δ_{max}	Maximum Axial Compression
n_y	Factor of Safety (FOS)
k	Axial Stiffness
M	Moment
m	Mass
ν	Poisson Ratio
v	Velocity
σ	Stress
L	Length
I	Second Area Moment of Inertia
S_Y	Yielding Strength
ρ	Density

1. INTRODUCTION

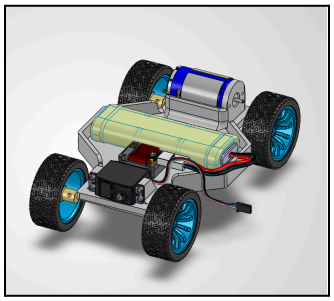


FIGURE 1: INITIAL OVERALL ASSEMBLY

Engineering is studied via learning theory, then applying it. University is optimal for this, as a course can teach a concept and provide the resources to apply it to a project. This is what our course has elected to do. Thus far, the main content of the course has been stress analysis and how stress affects materials of different shapes and sizes. We have analyzed how forces propagate stress through complex shapes, found points on a system that represent the maximum stress it withstands, and understood how different geometric features concentrate stress. Building an RC car allows us to show the concepts we learn in class hands-on. Additionally, this project enables direct testing against other teams to strengthen our learning through competition. The race tests each car's speed, handling, and endurance, exposing which teams most optimally designed their car. Win or lose, we will be satisfied to test engineering theory in an environment that simulated the life-cycle of an engineering product.

2. MATERIALS AND METHODS

In determining the most suitable material and method, we determined the overall goals, then analyzed possible materials based on parameters determined to optimize the chassis. Each parameter is explored through the Measure of Effectiveness (MOE) and Measure of Performance (MOP). Both conclusions were then analyzed to determine a final method and material.

2.1 Goals & Constraints

The goal is to design and fabricate a single-body, uni-material RC car chassis optimized to be lightweight and durable, surviving on rough terrain. The method and material limitations were defined by the resources available at Texas Inventionworks (TIW). Per the constraints, five candidates across three categories were determined: polymers, wood, and metal. These include the limitations of manufacturing methods at TIW: 3D printing, laser cutting, and machining.

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2.2 Framework

To evaluate these materials, the goals were broken down into three MOEs: agility, durability, and manufacturability. These goals were then associated with MOPs: agility with mass and Young's modulus, durability with stiffness and impact strength, and manufacturability with cost, fabrication time, ease of rapid iteration, and method of manufacturing.

2.3 Initial Screening

To narrow candidates before detailed analysis, materials were first screened for durability. Acrylic was eliminated first. Although the tensile strength of 55 MPa and modulus of 2.4-3.4 GPa are comparable to PLA and PETG, acrylic exhibits brittle characteristics with a notched Izod impact strength of 0.4-0.5 ft·lb/in with an elongation at break of only 2.7% [1], inferior to PLA and PETG, which exhibit elongations of 6.3% and 8.4%, respectively [2,3]. Due to the race being on rocky terrain and the risk of a crash, a material that fractures easily isn't suitable.

Baltic birch plywood was eliminated as it is highly anisotropic, with a bending modulus of 10.5 GPa with the face grain but only 3.2 GPa normal. The bending strength shows similar behavior, with a change from 114.3 MPa to 48.5 MPa [4]. Kallakas et al. [4] tested 9 mm 7-ply birch plywood, which could potentially differ from the material we would use, but the anisotropic behavior cannot be ignored. Although we could orient the plywood to exhibit the strongest behavior parallel to an oncoming impact force, the unknown direction of the force leaves other areas vulnerable to failure.

Both materials require laser cutting, which limits our material to a flat profile, limiting the methods of securing components such as the servo or drive motor to the chassis.

2.4 Detailed Analysis

The remaining materials are PLA, PETG, and 6061 Aluminum. Their material properties are:

	PLA	PETG	6061-T6
Density, ρ (g/cm³)	1.17 [2]	1.25 [3]	2.70 [5]
Young's Modulus (GPa)	3.43 [2]	2.12 [3]	68.9 [5]
Elongation at break (%)	6.3 [2]	8.4 [3]	12-17 [5]
Notched Charpy impact (kJ/m²)	3.3 [2]	2.6 [3]	-

TABLE 1: MATERIAL PROPERTIES OF PLA PETG AND 6061-T6 ALUMINUM

The properties of PLA and PETG rely on printer settings. In Table 1, the PLA and PETG were measured on specimens with 100% infill, 2 shells, and 3 top/bottom layers.

To determine an ideal material, the first analysis was done on the density. 6061-T6 aluminum has a density of 2.70 g/cm³ [5], which is 2.3 times the density of PLA and 2.2 times the density of PETG. So the mass of an aluminum chassis will be significantly heavier than its PLA or PETG counterpart. Mass was weighted above other material properties for two reasons. One, the components are assigned so their mass is out of our control, but the chassis' isn't. Due to Newton's Second Law, mass also has a significant effect on acceleration. In addition, aluminum also has a much higher Young's Modulus, thus a higher overall stiffness. Increased mass and stiffness mean a larger impact force. Second, aluminum is less manufacturable. Regarding cost, a 1/4" 6061-T651 plate costs approximately $0.51 per square inch [7], compared to around $0.10 per square inch for a fully solid printed plate of the same thickness as PLA and PETG [8]. The fabrication time would also be higher, as PLA and PETG just require the time to slice the part to G-code and send it off to the printer, but aluminum would require CAM and operating a CNC. Since we could leave a print unattended, this also improves our ability to rapidly prototype the chassis.

PLA and PETG were compared across agility and durability, since both use FDM 3D printing as the method of manufacturing. While PLA is slightly lighter, it is much stiffer and has a slightly higher notched Charpy impact strength. PETG has a higher elongation at break, meaning it deforms more before fracturing [2,3]. PETG's lower stiffness also reduces the peak force in an impact.

2.5 Conclusion

PETG was selected due to its higher ductility and lower impact force, beneficial if a crash occurred on rocky terrain, outweighing the advantages PLA exhibits in terms of mass.

3. RESULTS AND DISCUSSION

3.1 Weight, Acceleration, and Speed Calculations

The total weight was determined by adding the volume of each section of the chassis together and multiplying by the density of PETG. Once the weight of the chassis was found, we added the known weight of all of our current components and, combined with the chassis weight, found the curb weight of the car to be 2.36 lbs (1.07 kgs). To find the max acceleration, we first computed the tractive effort provided by the wheels at the highest motor rotation speed. Factoring in rolling resistance, we found the max acceleration of the car to be 3.15 m/s². Lastly, the RC car top speed was found by taking the max speed of the motor and converting it directly to wheel speed via the driveshaft. This came out to be 17.4 mph (7.78 m/s).

3.2 Chassis Deflection Calculations

To calculate the deflection of the chassis, we set up an energy balance equation to model a collision. Assuming all kinetic energy from the collision was transferred into the chassis, we equated the kinetic energy of the car in motion to the compression energy of the chassis, modeling it as a spring:

$$KE = \frac{1}{2}mv^2 \quad (1)$$

$$U = \frac{1}{2}k\delta^2 \quad (2)$$

$$\frac{1}{2}mv^2 = \frac{1}{2}k\delta^2 \quad (3)$$

We could calculate the stiffness constant of the chassis using the following equation:

$$k = \frac{EA}{L} \quad (4)$$

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From there, we could calculate the maximum deflection expected from the chassis as:

$$\delta_{max} = v_{max}\cos(\theta)\sqrt{\frac{m}{k}} \quad (5)$$

The calculated axial stiffness of the chassis was 978,042.89 N/m. At an impact speed of 7.78 m/s, the elastic spring model predicted a maximum compression of 8.14 mm and a peak force of 7.957 kN. The corresponding nominal compressive stress was 16.44 MPa, giving a factor of safety of 0.69 relative to the reported compressive strength of 11.29 MPa for 25% infill 3D-printed PETG [9]. At an impact angle of 60° from the wall normal, the predicted compression and force decreased to 4.07 mm and 3.98 kN, respectively. These estimates assume elastic axial compression and neglect buckling, bending, and other energy-absorption mechanisms.

3.3 FEA Analysis

A preliminary static finite element analysis was conducted in SOLIDWORKS to examine the chassis response to the estimated maximum impact force of 7.957 kN. The analysis evaluated stress concentrations and deformation in the detailed chassis geometry and provided a comparison with the simplified axial-compression calculation in Section 3.2.

The bare chassis was modeled. A total force of 7.957 kN was distributed over the front end face and directed longitudinally toward the rear. The rear end face of the base plate was fixed against translation. This condition represents an idealized compression test and does not reproduce the inertial response of a freely moving vehicle during a collision.

3.3.1 Modeling Assumptions

The chassis was approximated as a homogeneous, isotropic, linear-elastic solid using properties: An effective Young's modulus of 320 MPa, per our hand calculations; a Poisson's ratio of 0.3; and a shear modulus of 115.9 MPa, from:

$$G = \frac{E}{2(1+\nu)} \quad (6)$$

And an effective density of 550 kg/m³, an estimate for 25% infill (assumed solid skins). Additional assumptions include: homogenized infill, print-direction effects are neglected, the chassis is one continuous body, static loading, equipment and running gear are omitted, the rear restraint is idealized for the compression test, plasticity, fracture, and layer separation, and geometric nonlinearity are excluded. Also, density does not affect this force-only static analysis when gravity and inertial loading are absent. Preliminary results are below. Displacement extremes are local and may occur at different nodes.

3.3.2 FEA Results: Chassis Compression

Quantity	Result
Max von Mises stress	38.78 MPa
Min longitudinal displacement, UZ	−10.17 mm
Min vertical displacement, UY	−41.20 mm
Max resultant displacement, URES	42.40 mm
Max equivalent strain	0.09277

TABLE 2: PRELIMINARY RESULTS

The stress plot indicates a localized concentration near the front sidewall-to-base transition.

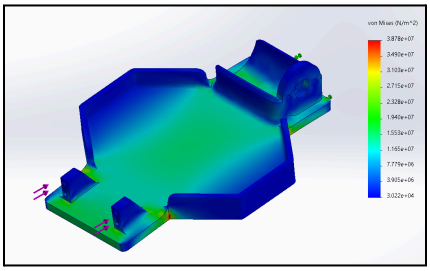


FIGURE 2: VON MISES STRESS UNDER COMPRESSION LOAD

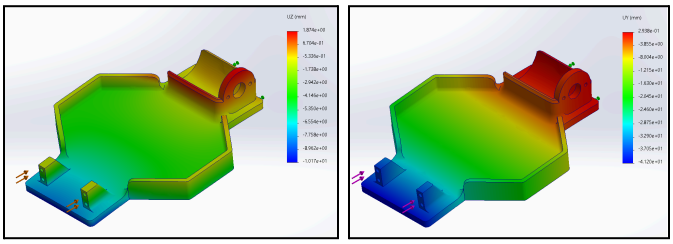


FIGURE 3 & 4: DISPLACEMENTS UZ (LEFT) AND UY (RIGHT)

The maximum longitudinal FEA displacement of 10.17 mm was 25% greater than the calculated axial compression of 8.14 mm, consistent with additional bending and local deformation in the detailed chassis model.

3.3.3 FEA Results: Angled Oblique Impacts

A preliminary 45° oblique static analysis was performed using longitudinal and lateral force components of 3,978.5 N each, producing a resultant of 5,626.5 N. Both components were distributed over the front end face while the rear base-plate end face remained fixed. The analysis predicted a maximum von Mises stress of 213.4 MPa and a maximum resultant displacement of 149.5 mm. The increased response relative to head-on loading suggests bending under lateral loading. However, the displacement and equivalent strain indicate that the linear-elastic results require nonlinear reassessment and cannot establish crash survivability.

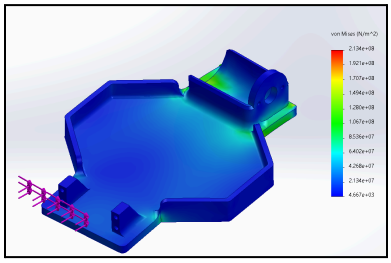


FIGURE 5: MAXIMUM VON MISES STRESS, 45° OBLIQUE LOAD

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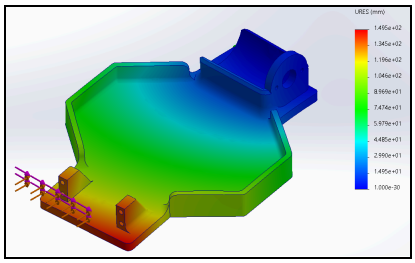


FIGURE 6: RESULTANT DISPLACEMENT, 45° OBLIQUE LOAD

The hand model predicts an axial compression of 5.76 mm at 45°, whereas FEA gives a resultant displacement of 149.5 mm; these measure different responses, with the FEA deformation indicating that nonlinear reassessment is needed.

Max result	Head-on	Revised 45°
Von Mises stress	38.78 MPa	213.4 MPa
Resultant displacement	42.40 mm	149.5 mm
Equivalent strain	0.09277	0.5141 (51.41%)

TABLE 3: COMPARISON TO HEAD-ON RESULTS

3.3.4 FEA Results: Sidewall Compressions

The exterior face of the opposite sidewall was fixed to provide the balancing reaction and prevent rigid-body motion during an idealized compression test, not to represent actual crash support. Simulations were run for each of the 3 faces.

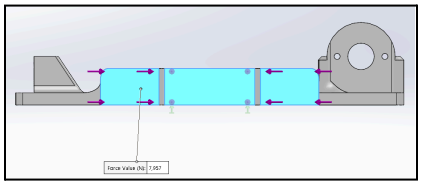


FIGURE 7: SIDE WALLS WHERE FORCES WERE APPLIED INDIVIDUALLY

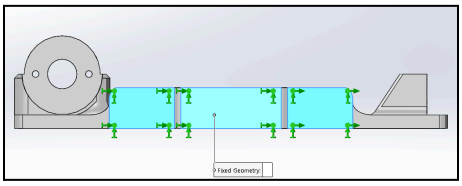


FIGURE 8: OPPOSITE SIDE, MODELED AS FIXED

Loaded sidewall location	Max von Mises stress (MPa)	Max resultant displacement (mm)	Max equivalent strain
1: Middle wall	342.1	544.9	0.8085
2: Rear wall	418.2	237.8	0.9951
3: Front wall	510.4	269.3	0.9561

TABLE 4: SIDEWALL IMPACT RESULTS

Preliminary linear-elastic outputs for idealized compression tests; the large displacements and strains exceed the model's valid predictive range. The front produced the highest peak stress, while the middle produced the largest resultant displacement. These differences reflect changes in load location, direction, and restraint location. All cases produced deformation beyond the useful range of the linear-elastic approximation; interpret values as screening outputs.

3.3.5 Drive Shaft/Axle

The drive shaft was modeled as a solid circular steel shaft with a diameter of 4.5 mm, to have direct impact with the wall. An overhang of 5 mm was assumed. Using the maximum impact force from the chassis, the impact is assumed to be divided equally between the wheels for a symmetric collision. Where the wheel force creates a bending moment at the bearing

$$F_w = \frac{F_{max} cos(\theta)}{2} \quad (7)$$

The moment was used to calculate shaft bending stress and elastic deflection, assuming the elastic modulus of our steel is 200 GPa, allowing for the calculation of the factor of safety.

$$M = F_w L_o \quad (7)$$

$$I = \frac{\pi d^4}{64} \quad (8)$$

$$\sigma_b = \frac{32M}{\pi d^3} \quad (9)$$

$$\delta = \frac{F_w L_o^3}{3EI} \quad (10)$$

$$n_y = \frac{S_y}{\sigma_b} \quad (11)$$

The wheel force was calculated to be 3978.5 N, giving us a bending moment of 19.89 N-m; the stress was found to be 2223 MPa; finding the area moment of inertia of 20.13 mm^4 yields a deflection of 0.0412 mm, and taking the factor of safety shows that 0.112 < 1, predicting yielding. Accounting for angles of 45° and 60° would still predict yielding. These calculations neglect drivetrain torque, unequal wheel loading, exact steel grade, and other energy-absorption mechanisms.

1. CONCLUSION

We selected our materials for our RC car chassis and ran initial testing. PETG was selected as chassis material based on its mass, ductility, and ease of printing. Hand calculations estimated a top speed of 7.78 m/s and material yielding as a result of the peak direct impact force of 7.957 kN, suggesting poor compressive strength. Preliminary FEA identified additional bending and stress concentrations under oblique and sidewall loading. Further refinement, nonlinear analysis, and testing should be done before confirming crash resistance.

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APPENDIX

Calculations

Page 1: PETG board

PETG
Density: 1.25 g/cm³ → .04516 lbf/in³

Board Weight
ρV = .04516 · 11.615 = **.525 lbs**

Board dimensions (sketch): top width 3, top section 1.75, side 4.5, overall length 8, wide section 6, angled section 2, small chamfer dims 1.95 / 1.25 / 1.5. Thickness: .25 in.
Motor mount cutout: 1.5 in, 2.49 × 1.76

Wall (dims 1.95, 2, .5; thickness .15 in)
Wall Area = 2(.5)·2 + 1.95(.5)·4 = 5.9 in²
Wall Volume = .885 in³

Board Volume
.25(10.5 + 13.5 + 9.76) + 2.29 + .885 = **11.615 in³**

Areas:

- Rectangle 3 × 1.75: A = 5.25 in² ×2 = 10.5 in²
- Rectangle 3 × 4.5: A = 13.5 in²
- Trapezoid (2 in top, 4.5 in bottom, 1.5 in tall): A = 4.88 in² ×2 = 9.76 in²

Motor mount:

- A = 4.38 in² (2.49 × 1.76), thickness 1.5 in
- 4.38 in² · 1.5 in = 6.57 in³
- Circular cut radius: .74 in
- Circular cut volume = πr²h = π(.74)²(2.49) = 4.28 in³
- Motor mount volume: 6.57 − 4.28 = **2.29 in³**

Page 2: Part weights

Part	Mass	Weight
Battery	360 g	.79 lbs
Servo	37 g	.082 lbs
ESC	39 g	.086 lbs
Motor	168 g	.386 lbs
Receiver	6 g	.013 lbs
Wheels	138 g	.304 lbs

Axle Gear: ρ = .283 lb/in³
V = πR²h + πr²h − πr_b²h
= π(1.22/2)²(.2) + π(1.1/2)²(.47 − .2) − π(.24/2)²(.47) = .47 in³
W = .47 · .283 = **.13 lbs**

Motor Gear: mod = d/z, d = mod·z = .5(12) = 6 mm, h ≈ 7 mm
πR²h − πr_b²h = π(6)²(7) − π(3)²(6) = 622 mm³ = .037 in³
ρ = .282 lb/in³, W = .037 · .282 = **.010 lb**

Driveshaft:
πR²h = π(.098)²(3.94) = .118 in³
ρ = .282, W = .118 · .282 = **.03 lb**

Part Total: 1.834 lbs
Car Weight: 2.36 lbs (boxed) | **Car Mass: 1.07 kg**

Page 3: Top speed and max acceleration

Top Speed
Motor max ω = 23,400 RPM
Add gearing: 23,400 / 4 = 5850 RPM @ wheels
Convert: 5850 · 2π = 36756.63 rad/min = 2205398.043 rad/h
Linearize: V = ωr = 2205398.043 rad/h · .0508 m = 112034.2206 m/h

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Convert to MPH / account for efficiency: 112034 · .25 / 1609 = **17.4 MPH** (boxed)

Maximum Acceleration
Limiting factor?
ESC: 60 A cont. ✓
Motor stall: 70 A

Estimate torque constant (stall / max efficiency)
K_t = τ/I = 196E-3 / 70 = .0028
30.6 / 13 = .00235
K_t = (.0028 + .00235) / 2 = .0026

Estimate torque at 60 A
τ = K_t · I = .0026 · 60 = .15 N·m

Tractive Effort
(Assume wheel radius 2 in ≈ .0508 m and gear ratio 4:1, ignore drag)
F = τ · η_e · N_tf / r = .15 · .3 · 4 / .0508 = **3.54 N**

Page 4: Acceleration

Motor speed @ 60 A: 5000 RPM

Convert to wheel speed:
5000 · 2π · 60 · .0508 · .28 / (4 · 1609) = **4.17 MPH**

Rolling Resistance (Gillespie's Vehicle Dynamics textbook)
f_r = f_0 + 3.24 f_s (V/100)^2.5, @ 4.2 mph (60 A draw)
f_r = .02 + 3.24(.02)(4.2/100)^2.5 = .02

R_x = f_r · W
R_x = .02(8.39) = .17 N

Acceleration
ΣF = ma
a = (F − F_A)/m = (3.54 − .17) / 1.07 = **3.15 m/s²** (boxed)

Chassis Deflection Calculations

$KE = \frac{1}{2}mv^2$

$U = \frac{1}{2}k\delta^2$

$KE = U = \frac{1}{2}mv^2 = \frac{1}{2}k\delta^2$

$\frac{1}{2}m(v\cos(\theta))^2 = \frac{1}{2}k\delta^2$

$\delta_{max} = v_{max}\cos(\theta)\sqrt{\frac{m}{k}}$

$k = \frac{EA}{L}$

$k_i = \frac{EA_i}{L_i}$

$F_{collision} = k_{total}\delta_{max} = \sum_{i=1}^{5} k_i\delta_i = \sum_{i=1}^{5} \frac{EA_i}{L_i} = \sum_{i=1}^{5} \frac{Etb_i(x)}{L_i}$

Section the chassis into 5 parts to calculate the area:

Section I (constant width)
$b_1(x) = 3\,\{0 \le x \le 1.75\}$

Section II (non-constant width)
$b_2(x) = 3 + \frac{6-3}{1.25}(x - 1.75)\,\{1.75 \le x \le 3\}$

Section III (constant width)
$b_3(x) = 6\,\{3 \le x \le 5\}$

Section IV (non-constant width)
$b_4(x) = 6 + \frac{3-6}{1.25}(x - 5)\,\{5 \le x \le 6.25\}$

Section V (constant width)
$b_5(x) = 3\,\{6.25 \le x \le 8\}$

$d\delta = \frac{Fdx}{EA(x)} \Rightarrow \delta = \frac{F}{Et}\int_0^L \frac{dx}{b(x)} \Rightarrow \frac{\delta}{F} = \frac{1}{Et}\int_0^L \frac{dx}{b(x)}$

$F = k\delta \Rightarrow \frac{1}{k} = \frac{\delta}{F}$

$\frac{1}{k} = \frac{1}{Et}\int_0^L \frac{dx}{b(x)}$

To convert from inches to m, divide the length in inches by 39.37

$\frac{1}{k} = \frac{1}{0.32\cdot 10^9\left(\frac{0.25}{39.37}\right)}\left(\int_0^{1.75}\frac{1}{b_1}dx + \int_{1.75}^{3}\frac{1}{b_2}dx + \int_3^5\frac{1}{b_3}dx + \int_5^{6.25}\frac{1}{b_4}dx + \int_{6.25}^{8}\frac{1}{b_5}dx\right)$

$k = 978042.891259\ \frac{N}{m}$

$\delta_{max} = v_{max}\cos(\theta)\sqrt{\frac{m}{k}}$

$F_{max} = k\delta_{max}$

$\sigma = \frac{F_{max}}{A_{min}}$

At $\theta = 0°$

$\delta_{max} = 7.78\frac{m}{s}\sqrt{\frac{1.07kg}{978042.891259\frac{N}{m}}}$

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$\delta_{max} \approx 8.14\,mm$

$F_{max} = k\delta_{max} = 978042.891259\,\frac{N}{m} \cdot 0.00814m = 7956.81N$

$\sigma_{max} = \frac{F_{max}}{A_{min}} = \frac{7956.81}{(\frac{3 \cdot 0.25}{1550})} = 16.44\,MPa$

The literature value for the factor of safety of PETG under compression is 11. 29MPa

Calculating the yield factor of safety results in

$n_y = \frac{11.29}{16.44} = 0.69$ (we predict yielding)

At $\theta = 45°$
$\delta_{max} \approx 5.75\,mm$
$F_{max} = k\delta_{max} = 978042.891259\,\frac{N}{m} \cdot 0.00575m = 5626.31N$
$\sigma_{max} = \frac{F_{max}}{A_{min}} = \frac{5626.31}{(\frac{3 \cdot 0.25}{1550})} = 11.63\,MPa$
$n_y = \frac{11.29}{11.63} = 0.97$ (we predict yielding)

At $\theta = 60°$
$\delta_{max} \approx 4.07\,mm$
$F_{max} = k\delta_{max} = 978042.891259\,\frac{N}{m} \cdot 0.00407m = 3978.40N$
$\sigma_{max} = \frac{F_{max}}{A_{min}} = \frac{3978.40}{(\frac{3 \cdot 0.25}{1550})} = 8.22\,MPa$
$n_y = \frac{11.29}{8.22} = 1.37$ (we don't predict yielding)

Drive Shaft/Axle

Shaft Dimensions:
$d = 4.5\,mm = 0.0045\,m$
$L_{shaft} = 3.94\,in = 100.1\,mm$
$L_o = 0.0045\,m$ (Overhang of wheel load to nearest support)
$E_{steel} = 200\,GPa$ (Assumed)
$S_y = 250\,MPa$ (Assumed)

Taking a head-on collision assumption (as well as not accounting for torque):
$F_w = \frac{F_{max}cos(\theta)}{2} = \frac{7.957cos(0)}{2} = 3.9785\,kN$
$M = F_w L_o = 3.9785 * 0.005 = 19.89\,N * m$
$I = \frac{\pi d^4}{64} = \frac{\pi(0.0045)^4}{64} = 2.013 * 10^{-11} m^4$
$\sigma_b = \frac{32M}{\pi d^3} = \frac{32(19.89)}{\pi(0.0045)} = 2223\,MPa$

$\delta = \frac{F_w L_o^3}{3EI} = \frac{3978.5(0.0045)^3}{3(200*10^9)(2.013*10^{-11})} = 0.015\,mm$
$n_y = \frac{S_y}{\sigma_b} = \frac{250}{2223} = 0.112 < 1$ (Yields)
45° - $n_y = 0.159$ (Yields)
60° - $n_y = 0.225$ (Yields)

The tolerances used for screw holes were 0.0125, used for both the servo mounting and DC mounting holes.

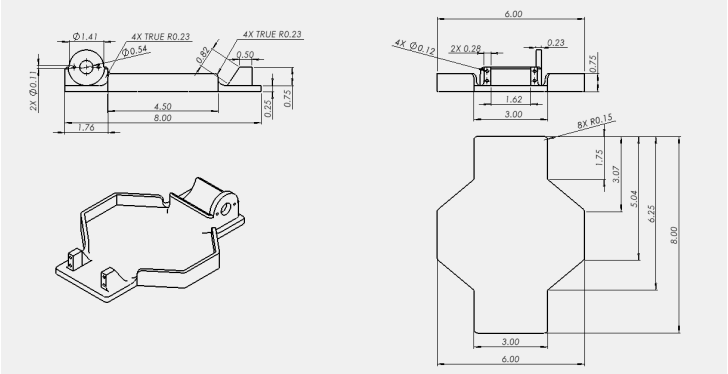


FIGURE 10: ENGINEERING DRAWING OF CHASSIS CAD

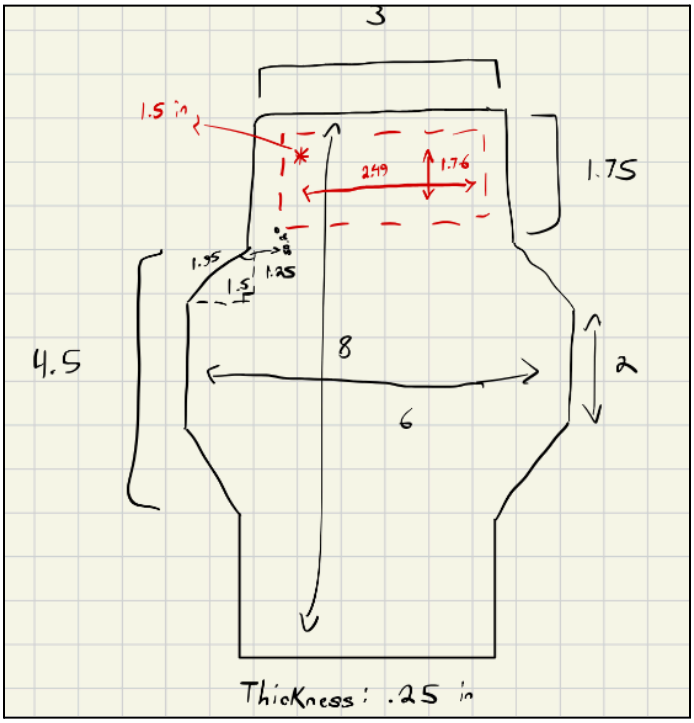


FIGURE 11: INITIAL SKETCH OF CHASSIS BODY

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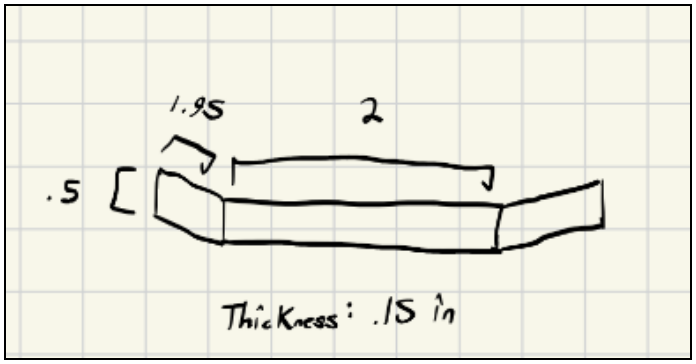


FIGURE 12: INITIAL SKETCH OF CHASSIS WALL BRACKETS

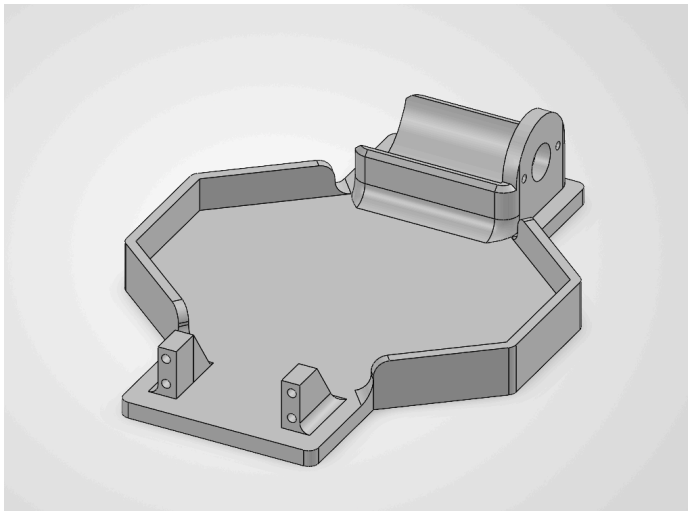


FIGURE 13: INITIAL CAD OF CHASSIS

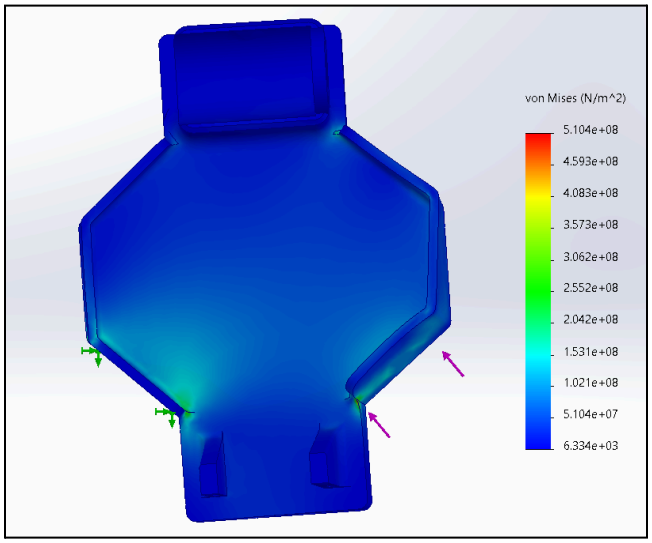


FIGURE 14: FEA, VON MISES STRESS ON A WALL

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